CHAPTER 1: PHOTOSYNTHESIS OVERVIEW

Photosynthesis is the fundamental biological process by which green plants, algae, and cyanobacteria convert light energy into chemical energy.

This chemical energy is stored in carbohydrate molecules such as glucose.

The chemical equation for oxygenic photosynthesis is:

$6CO_2 + 6H_2O + \text{Light Energy} \rightarrow C_6H_{12}O_6 + 6O_2$.

The green pigment chlorophyll absorbs blue and red wavelengths of light.

CHAPTER 2: LIGHT REACTIONS AND THYLAKOID MEMBRANE

The light-dependent reactions take place on the thylakoid membrane inside chloroplasts. During this stage, photons strike photosystem II, splitting water molecules through photolysis. This releases protons, electrons, and free oxygen gas as a byproduct.

Electrons move through an electron transport chain to produce ATP and NADPH energy carriers.

CHAPTER 3: CALVIN CYCLE AND CARBON FIXATION

The Calvin cycle takes place in the stroma of the chloroplast.

It uses the ATP and NADPH synthesized in the light reactions to convert CO2 into glucose.

The enzyme RuBisCO catalyzes the initial carbon fixation step with ribulose 1,5-bisphosphate.

Temperature, light intensity, and CO2 concentration are the primary factors affecting the rate.